

HDOC DAY-31 ASSIGNMENT

Menu-Driven Operations on a 2D Array

1. Problem Understanding

The task is to write a menu-driven C program for a given 2D integer array. The user enters the number of rows and columns and then enters all matrix elements. A switch statement is used to perform the selected operation.

- Find the sum of all elements in the matrix.
- Find the sum of each row and store the row sums in a 1D array.
- Find and display the transpose of the matrix.
- Exit the program.

2. Standard Test Cases

The test cases include every input required by the program: number of rows, number of columns, complete matrix elements, and menu choice.

Test Case	Rows	Columns	Matrix Elements	Choice	Expected Output	Status
TC1	2	3	1 2 3 / 4 5 6	1	Sum of all elements = 21	Pass
TC2	2	3	1 2 3 / 4 5 6	2	Row sums: 6, 15	Pass
TC3	2	3	1 2 3 / 4 5 6	3	Transpose: 1 4 / 2 5 / 3 6	Pass
TC4	2	2	2 4 / 6 8	1	Sum of all elements = 20	Pass
TC5	2	2	2 4 / 6 8	2	Row sums: 6, 14	Pass
TC6	2	2	2 4 / 6 8	3	Transpose: 2 6 / 4 8	Pass
TC7	2	3	1 2 3 / 4 5 6	4	Program exits	Pass

3. Algorithm

A. Common Menu Logic

1. Start.
2. Read the number of rows and columns.
3. Read all elements of the matrix using nested loops.
4. Display the menu options.
5. Read the user's choice.
6. Use switch-case to perform the selected operation.
7. Repeat the menu until the user chooses Exit.
8. Stop.

B. Sum of All Elements

1. Set sum = 0.
2. Visit every matrix element using nested loops.
3. Add each element to sum.
4. Display sum.

C. Sum of Each Row

1. For each row, set rowSum = 0.
2. Add all elements of that row to rowSum.
3. Store rowSum in the row-sum array.
4. Display each row sum and the complete row-sum array.

D. Transpose of the Matrix

1. Use the column index as the outer loop.
2. Use the row index as the inner loop.
3. Display matrix[i][j] as matrix[j][i] by printing matrix[j][i] order through the loops.
4. Move to the next line after each transposed row.

4. C Program

```
#include <stdio.h>

int main()
{
    int rows, columns, i, j, choice;
    int matrix[10][10];
    int rowSum[10];
    int sum;

    printf("Enter the number of rows and columns: ");
    scanf("%d %d", &rows, &columns);
    printf("Enter the matrix elements:\n");
    for (i = 0; i < rows; i++)
    {
        for (j = 0; j < columns; j++)
        {
            scanf("%d", &matrix[i][j]);
        }
    }
    do
    {
        printf("\nMENU\n");
        printf("1. Sum of all elements\n");
        printf("2. Sum of each row\n");
        printf("3. Transpose of matrix\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice)
        {
            case 1:
                sum = 0;
```

```

for (i = 0; i < rows; i++)
{
    for (j = 0; j < columns; j++)
    {
        sum = sum + matrix[i][j];
    }
}
printf("Sum of all elements = %d\n", sum);
break;

```

case 2:

```

printf("Row sums:\n");
for (i = 0; i < rows; i++)
{
    rowSum[i] = 0;
    for (j = 0; j < columns; j++)
    {
        rowSum[i] = rowSum[i] + matrix[i][j];
    }
    printf("Sum of row %d = %d\n", i + 1, rowSum[i]);
}
printf("The row-sum array contains: [");
for (i = 0; i < rows; i++)
{
    printf("%d", rowSum[i]);
    if (i < rows - 1)
    {
        printf(", ");
    }
}
printf("]\n");
break;

```

case 3:

```
printf("Transpose of the matrix:\n");
```

```
for (j = 0; j < columns; j++)
```

```
{
```

```
    for (i = 0; i < rows; i++)
```

```
    {
```

```
        printf("%d\t", matrix[i][j]);
```

```
    }
```

```
    printf("\n");
```

```
}
```

```
break;
```

case 4:

```
printf("Program exited.\n");
```

```
break;
```

default:

```
printf("Invalid choice.\n");
```

```
}
```

```
}
```

```
while (choice != 4);
```

```
return 0;
```

```
}
```

5. Compilation and Execution Evidence

```
kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
└─$ gedit question.c

** (gedit:2070): WARNING **: 10:29:23.838: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:2070): WARNING **: 10:29:23.838: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

(kali@kali)-[~/Desktop]
└─$ gcc question.c -o question

(kali@kali)-[~/Desktop]
└─$ ./question
Enter the number of rows and columns: 2
3
Enter the matrix elements:
1
2
3
4
5
6
C
MENU
1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit
Enter your choice: 1
Sum of all elements = 21

MENU
1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit
Enter your choice: 2
Row sums:
Sum of row 1 = 6
Sum of row 2 = 15
The row-sum array contains: [6, 15]
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help

Row sums:
Sum of row 1 = 6
Sum of row 2 = 15
The row-sum array contains: [6, 15]

MENU
1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit
Enter your choice: 3
Transpose of the matrix:
1      4
2      5
3      6

MENU
1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit
Enter your choice: 4
Program exited.

(kali@kali)-[~/Desktop]
└─$ ./question
Enter the number of rows and columns: 2
2
Enter the matrix elements:
2
4
6
8
MENU
1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit
Enter your choice: 1
Sum of all elements = 20
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help

MENU
1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit
Enter your choice: 1
Sum of all elements = 20

MENU
1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit
Enter your choice: 2
Row sums:
Sum of row 1 = 6
Sum of row 2 = 14
The row-sum array contains: [6, 14]

MENU
1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit
Enter your choice: 3
Transpose of the matrix:
2      6
4      8

MENU
1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit
Enter your choice: 4
Program exited.

(kali@kali)-[~/Desktop]
```

6. Verification of All Test Cases

Test Case	Row s	Cols	Matrix Elements	Choice	Expected Result	Actual Result	Status
TC1	2	3	1 2 3 / 4 5 6	1	21	21	Pass
TC2	2	3	1 2 3 / 4 5 6	2	6, 15	6, 15	Pass
TC3	2	3	1 2 3 / 4 5 6	3	1 4 / 2 5 / 3 6	1 4 / 2 5 / 3 6	Pass
TC4	2	2	2 4 / 6 8	1	20	20	Pass
TC5	2	2	2 4 / 6 8	2	6, 14	6, 14	Pass
TC6	2	2	2 4 / 6 8	3	2 6 / 4 8	2 6 / 4 8	Pass
TC7	2	3	1 2 3 / 4 5 6	4	Exit	Exit	Pass